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Compact and ultrastable photonic microwave oscillator

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Frequency comb synthesized microwaves have been so far realized with tabletop systems, operated in well-controlled environments. Here, we demonstrate state-of-the-art ultrastable microwave synthesis with a compact rack-mountable apparatus. We present absolute phase noise characterization of a 12 GHz signal using an ultrastable laser at ~ 194 THz and an Er:fiber comb divider, obtaining -83 dBc/Hz at 1 Hz and < -166 dBc/Hz for offsets greater than 5 kHz. Employing semiconductor coating mirrors for the same type of transportable optical frequency reference, we show that -105 dBc/Hz at 1 Hz is supported by demonstrating a residual noise limit of division and detection process of -115 dBc/Hz at 1 Hz. This level of fidelity paves the way for the deployment of ultrastable photonic microwave oscillators and for operating transportable optical clocks.

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Femtosecond laser frequency combs were envisioned to simplify the determination of optical frequencies with respect to radio-frequency (RF) standards [1,2]. Being a direct coherent link between the two frequency domains, the synthesis of radio and microwave signals with a stable phase relation with an optical frequency reference (OFR) is viable [3]. Photonic synthesis of ultrastable microwaves via phase-coherent division of optical signals is in fact becoming more relevant for high-performance radar systems [4], deep-space communication, high-speed analog-to-digital sampling, timing distribution, and synchronization for large-scale scientific experiments or very long baseline interferometry. Moreover, downconversion of optical frequency standards is a required process for the realization of optical clocks, their use as quantum sensors, and eventually for the realization of optical timekeeping and the redefinition of the SI-second [5]. The leading technologies to generate ultrastable microwaves are cryogenic (for close to carrier offset frequencies) [6] or room temperature sapphire-loaded cavity oscillators and opto-electronic oscillators (for high offset frequencies) [7,8]. However, phase-coherent division by frequency combs proved to rival established technologies for the whole Fourier frequency spectrum [9–13], reducing the required high maintenance of cryogenic systems. Furthermore, the comb division approach provides tunability on the synthesized frequency that is not restricted by the mechanical dimension of the sapphire-loaded cavities. With tremendous progress made on OFRs, megahertz-level linewidths [14] and 1×10^{-19}/s fractional linear drifts have been demonstrated [15]. On the other hand, progress on ultralow noise frequency comb control allows the transfer of $< 10^{-17}$-level stability to a broad range of comb modes [16,17]. Besides the necessity of an OFR with low phase noise, the comb requires the division of such spectral purity without additive noise. For an ideal optical frequency divider (OFD), the phase noise of the OFR is reduced to $S_{\varphi,\mu} = S_{\varphi,\nu}/n^2$, where $S_{\varphi,\nu}$ and $S_{\varphi,\mu,}$ respectively, are the optical and microwave phase noise power spectral densities (PSDs), and n is the ratio ($\sim 10^4$) between the optical (ν) and microwave carrier (μ) signals.

There is an increasing interest on porting the aforementioned performance on transportable systems; in fact, frequency combs can be constructed in a compact form factor with high robustness, and their operation on sounding rockets has been demonstrated. In addition, transportable OFRs achieving $< 10^{-15}$ level of stability at short time scales have been conceived [18,19].

Here, the characterization of a deployable photonic microwave oscillator (PMO) integrating a compact OFR and a fully stabilized ultralow noise OFD [20,21], yet demonstrating zeptosecond-level timing noise for a 12 GHz microwave signal, is carried out. Previously published results in Ref. [12] were

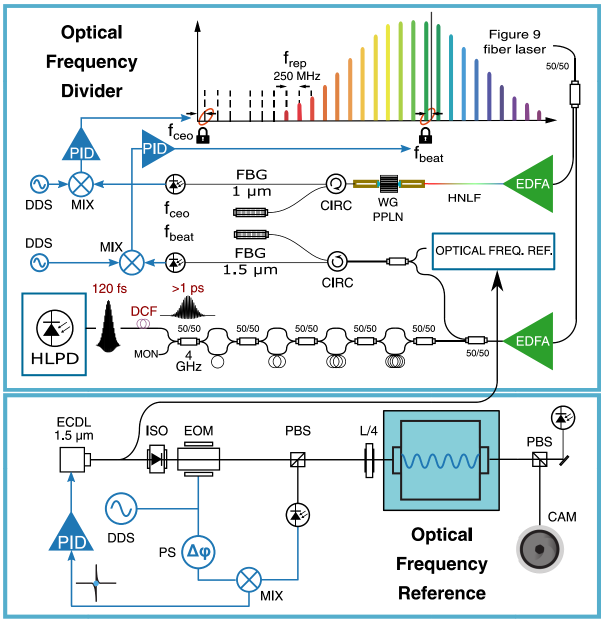

Fig. 1. PMO scheme. Top, the OFD; bottom, the OFR. f_{ceo}, carrier-to-envelope offset frequency; f_{rep}, repetition rate frequency; f_{beat}, heterodyne beat frequency; 50/50, fiber splitter; EDFA, Er-doped fiber amplifier; HNLF, highly nonlinear fiber; WG PPLN, waveguide periodically poled lithium niobate; CIRC, circulator; FBG, fiber Bragg grating; PID, proportional-integral-derivative controller; DDS, direct-digital-synthesizer; MIX, RF mixer; DCF, dispersion-compensating fiber; HLPD, highly linear photodetector; PS, RF phase-shifter; PBS, polarizing beam splitter; ISO, optical isolator; EOM, electro-optic modulator; L/4, quarter-wave retarder plate; CAM, CMOS IR camera.

realized employing laboratory-size setups and, for the comb's carrier-to-envelope offset frequency control, a hybrid approach of the transfer oscillator scheme [22] was implemented. That setup consisted of large apparatuses, making it impractical for many applications. Instead, here we demonstrate comparable spectral fidelity in a transportable and fully phase-locked apparatus, in some respects improving the performance and lowering the complexity. Such technology enables, to name a few, in-field operation of transportable optical lattice clocks, 5G and 6G communication, and vehicle-size radar systems.

A schematic of the main building blocks of the apparatus is shown in Fig. 1. The core of the OFD is a femtosecond laser oscillator, more specifically a figure 9 nonlinear amplifying loop mirror-based Er:fiber laser with a repetition rate of 250 MHz, centered at ~ 192 THz [23]. With approximately 250 mW of pump power at 980 nm, a 40 nm full width at half-maximum spectrum is obtained, and < 50 fs pulses are generated. At the fiber-coupled output of the laser, an average power of ~ 5 mW is available. The pulse train is moderately amplified in an Er-doped fiber amplifier (EDFA) and split into two branches. One is used for generating a supercontinuum in a highly nonlinear fiber spanning more than an octave which is subsequently used in an f-to-2f all-fiber interferometer for the detection of the CEO frequency. The octave spanning spectrum is coupled into a periodically-poled lithium niobate (PPLN) waveguide, and the 1f spectral components doubled from the crystal are beaten in a multi-line interferometer with the fundamental comb light at 2f, approximately having 3,000 comb lines filtered by a fiber Bragg grating (FBG) at ~ 1 µm. The second output of the oscillator is seeding a low relative intensity noise (RIN) EDFA, showing RIN of < -110 dB/Hz at 1 Hz, rolling off to < -138 dB/Hz at 10 kHz and < -160 dB/Hz at offset frequencies larger than 100 kHz. The optical referencing of the comb spacing to an OFR is possible via heterodyne beat detection with an integrated interferometer and photodiode unit. Upon detection of the two degrees of freedom, the laser intra-cavity actuators and the integrated fast locking electronics allow ~ 1.5 MHz bandwidth on the phase-locked loops. The inclusion of a frequency counter ensures their monitoring over time (details in [21]). The amplified 250 MHz pulse train is multiplied up to 4 GHz using a four-fold pulse interleaver based on cascaded Mach–Zehnder interferometers [24]. Due to the overall fiber length, a substantial dispersion is accumulated causing a pulse width broadening exceeding 1 ps. It has been shown that pulses below 1 ps are necessary for obtaining the lowest microwave phase noise. One crucial reason is that the total shot noise of the comb projects unevenly on amplitude and phase noise, with the latter being dramatically reduced when the pulse width is in the hundreds of femtosecond regime [25]. Therefore, we have implemented a dispersion-compensating fiber (DCF) to compress the pulses on average to a duration of 120 fs. Impinging the pulse train onto a highly linear dual-depletion InP/InGaAs PIN photodetector (HLPD, Discovery Semiconductors Inc.), the fundamental carrier at 4 GHz and its harmonics can be extracted, yielding 0 dBm at 12 GHz and a photocurrent of ~ 8 mA. The OFD is enclosed in 3 height-unit (3U) 19" rack-mounted housing with dimensions of $48 \times 56 \times 13\ \mathrm{cm}^3$ (width, depth, height) for a volume of 35 l, a weight of 20 kg, and power consumption of < 50 W. The comb control is fully accessible by an embedded system (further details in Ref. [21]). The phase noise of the in-loop phase-locked loops controlling the OFD for self- and optical referencing, are projected down to the 12 GHz carrier (subtracting 84 dB) and displayed in the red and green traces in Fig. 2. The residual phase noise of the 12 GHz signal has been determined via heterodyne digital cross correlation as performed in Ref. [12]. The division limit, set by the OFD and photodetection (PD), is obtained referencing three OFDs to the same OFR. Generating three microwave carriers at 11.995, 12, and 12.005 GHz, respectively, the two resulting 5 MHz signals obtained by mixing the signals

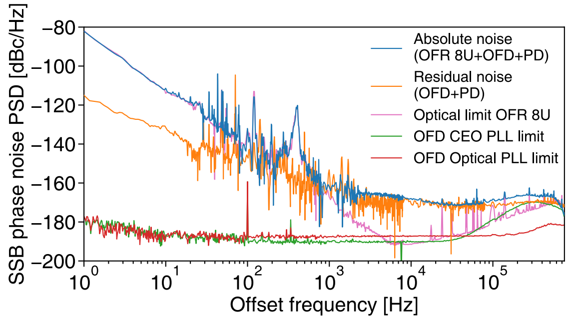

Fig. 2. Absolute phase noise of the compact PMO (blue). The residual noise of the comb division (OFD) and PD noise limit is shown in the orange trace. The limit inferred by the optical phase noise of the 8U OFR is shown in pink, while the green and red traces represent the projected in-loop phase noise of the phase-locked loops. The measurement noise floor does not limit the assessment [12].

are cross-correlated (while for the absolute phase noise three OFRs are used). The residual single-side band phase noise PSD of the photonic microwave reaches a level of -115 dBc/Hz at 1 Hz and < -168 dBc/Hz at 10 kHz, ensuring that the opto-electrical spectral purity transfer would be limited to $< 1 \times 10^{-16}$ at 1 s for a 12 GHz carrier signal.

The 8U 19" OFR at 194.4 THz features a Fabry–Perot optical cavity consisting of a 5 cm ultralow expansion glass (ULE) cube [26] (free-spectral range of 3 GHz) and ion-beam sputtered SiO_2/Ta_2O_5 coating mirrors on ULE substrate. The obtained finesse is $\sim 2.5 \times 10^5$, and the estimated thermal Brownian noise floor for the ULE cavity is $\sim 2 \times 10^{-15}$. A temperature stabilization loop maintains the cavity at its thermal expansion coefficient's zero-crossing. The vacuum chamber pressure is controlled by an ion getter pump maintaining a value $< 10^{-7}$ mbar. The chamber is equipped with two optical breadboards, one for hosting the in-coupling optics and, on the opposite side, one for supporting the transmission photodiode and a CMOS camera to monitor the TEM_{00} spatial profile. The vacuum and optics assembly are contained in a medium-density fiberboard box. A standard Pound–Drever–Hall locking scheme is implemented by generating 20 MHz side bands on the injected cavity light (from a RIO PLANEX CW fiber-coupled laser at 1,542.14 nm) yielding a bandwidth of ~ 600 kHz. The control of the OFR is fully accessible by an embedded system. Comprising the electronics, the overall dimensions are $48 \times 43 \times 35$ cm^3 for a volume of 72 l, 80 kg of weight, and consuming ~ 60 W.

To qualify the spectral fidelity of the 12 GHz signal, we have first measured the optical phase noise of the OFR. As described in Ref. [27], we have generated two optical heterodyne beat notes between the 8U OFR and two stationary OFRs to perform a digital cross correlation. In Fig. 2, this measurement is scaled down to a 12 GHz microwave carrier (subtracting 84 dB), shown by the pink trace, clearly limiting up to 1 kHz offset. The absolute phase noise of the 12 GHz carrier is displayed in blue, and it shows perfect agreement with the limit inferred by the optical phase noise measurement at a level of -83 dBc/(Hz $\times$ f^3) up to 1 kHz offset. The 400 Hz spike, appearing on the optical and microwave phase noise, is due to a resonance later identified within the OFR system. The absolute phase noise reaches values < -166 dBc/Hz for offsets greater than 5 kHz, maintaining this level for values > 100 kHz, resulting in a substantial improvement compared to our previous result [12]. Figure 3 shows the absolute phase noise spectrum from 1 kHz to 700 kHz and its integration in terms of timing jitter yields to 100 as, a value which is approximately 200 as lower than the one obtained in our previous work [12].

In the case of femtosecond pulse train PD, the peak intensity is far greater than for CW illumination. Saturation effects are prone to degrade the RF phase noise. The RIN of the laser light is then upconverted to the RF carrier degrading the opto-electrical phase transfer through the phenomenon known as amplitude-to-phase conversion (APC) [28,29]. For the PMO presented here, we have been operating the photodiode at a specific point in terms of optical power and reverse bias voltage to ensure 30 dB APC rejection without any active control, while a slow feedback loop [30] allowed a stable rejection up to 33 dB for the work in Ref. [12]. Here, at offset frequencies around 10 kHz, the measured phase noise PSD shows that the PMO is limited by the RIN of the optical pulses impinging on the photodiode (-138 dBc/Hz -30 dB $= -168$ dBc/Hz at 10 kHz).

In addition, we have optically qualified a 14U OFR also based on a 5 cm cubical cavity, but equipped with GaAs/AlGaAs crystalline coating mirrors (Crystalline Mirror Solutions Ltd.) on fused silica substrate [31] and ULE compensating rings [32]. The obtained finesse is $\sim 2.2 \times 10^5$, with an estimated thermal floor of $\sim 5 \times 10^{-16}$. The improved performance, shown in Figs. 4 and 5, in terms of stability and close-to-carrier phase noise, respectively, comes at the expense of a less compact OFR (62 cm higher, ~ 100 kg of weight). In order to operate it at 8×10^{-16} stability at a short time scale, additional optics, control electronics for intensity control of the CW laser, and

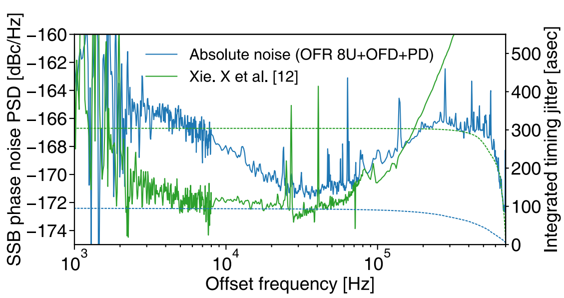

Fig. 3. Absolute phase noise for high offsets of the 12 GHz carrier of the results in Ref. [12] (green) and of the one reached by the compact PMO (blue). On the right, integrated timing jitter from 700 kHz down to 1 kHz.

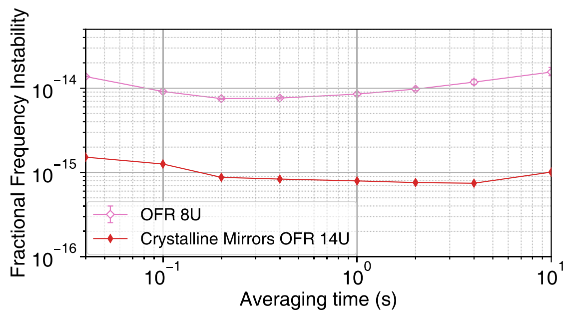

Fig. 4. Fractional frequency instability (modAllan deviation) for the 8U OFR (pink) and for the 14U OFR (red). The error bars, where visible, indicate 1 σ confidence intervals.

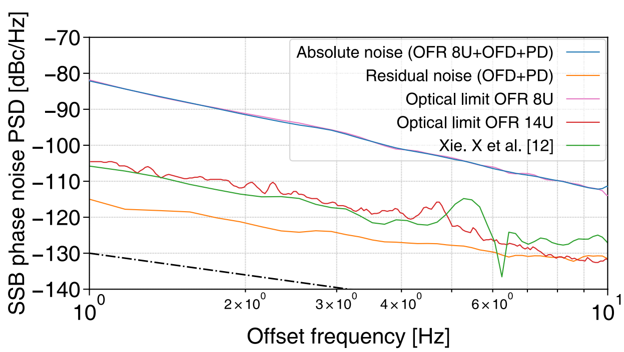

Fig. 5. Close to carrier absolute phase noise for the PMO (blue) and for results in Ref. [12] (green). The 8U (pink) and 14U (red) OFR optical limits and the residual comb division noise (orange) are shown (measurement noise floor, black dashed-dotted).

Doppler compensation of the fiber links for mitigating the residual amplitude modulation are required. Furthermore, inside the 14U chassis, a 2U active anti-vibration platform is installed. The overall power consumption of the 14U unit reaches up to ~ 90 W. The pink trace in Fig. 4 represents a measurement of the 8U OFR against a 1.5×10^{-15} at 1 s OFR, while the red trace is obtained by a three-cornered hat measurement. In Fig. 5, highlighting the close-to-carrier spectrum of the phase noise, we compare the inferred optical limit of the 14U OFR with the absolute phase noise assessed for the 8U OFR-based PMO and the previously reported performance of the tabletop apparatus. The residual phase noise of the division is lower than the optical limit set by the 14U OFR; therefore, the OFD can support this microwave spectral purity with negligible noise contribution.

Advances on PD technology, together with the implementation of an intensity noise servo loop for the OFD, are among the prospects for improvements of the PMO's performance described here. The OFRs could also serve to seed microcomb technologies [33], so far bound to laboratory operations due to the need for bulky setups.

In conclusion, the level of fidelity of such a versatile synthesizer represents a technological leap in microwave photonics, enabling in-field applications where the best in class microwave oscillators are required. Furthermore, the OFD, together with the 14U OFR, allows the operation of transportable optical clocks with stabilities below 10^{-15} level at 1 s, paving the way to sub-centimeter geodesy [34], and the realization of metrological networks for the generation and dissemination of an optical time scale [35].

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